Analysis of Algorithms

- Efficiency of an algorithm can be analyzed at two different stages, before implementation and after implementation.

I. A priori analysis

- Theoretical analysis of an algorithm.
- Efficiency of algorithm is measured by assuming that all other factors e.g. speed of processor, are constant and have no effect on implementation.

II. A posterior analysis

- Empirical analysis of an algorithm.
- The chosen algorithm is implemented using programming language.
- Next the chosen algorithm is executed on target computer machine.
- In this analysis, actual statistics like running time and space needed are collected.

- Algorithm analysis is dealt with the execution or running time of various operations involved.
- Running time of an operation can be defined as number of computer instructions executed per operation.

Priori & Posteriori Analysis-Difference

Priori	Posteriori
Analysis of an algorithm prior to running on any system	Analysis of an algorithm only after running on any specific system
Done on algorithms	Done on programs
Independent of language of compiler and types of hardware	Depend on language of compiler and types of hardware
It gives approximate values	Exact answer possible
Generally used as it gives same values for all systems	Not generally used,as a software is generally made for different users which uses different systems

Time & Space Complexity

- If the problem is having more than one solution or algorithm then the best one is decided by the analysis based on two factors.
    i. CPU Time (Time Complexity)
    ii. Main memory space (Space Complexity)

Time Complexity

✓ Time complexity of an algorithm quantifies the amount of time taken by an algorithm to run as a function of the length of the input.

Space Complexity

✓ Space complexity of an algorithm quantifies the amount of space or memory taken by an algorithm to run as a function of the length of the input.

- Time and space complexity depends on lots of things like hardware, operating system, processors, etc.
- However, we don't consider any of these factors while analyzing the algorithm.
- We will only consider the execution time of an algorithm.